

OSI (Open Systems Interconnection) Model – Simplified Table

Layer No.	Layer Name	Purpose (in Simple Words)	Example Protocols / Technologies	Real-Life Example
7	Application Layer	Directly interacts with user apps like browser, email, etc.	HTTP, FTP, SMTP	Opening a website or sending an email
6	Presentation Layer	Translates data, handles encryption & compression	SSL, TLS, JPEG, MPEG	Viewing a secure website or streaming a video
5	Session Layer	Manages start, use, and end of communication sessions	NetBIOS, RPC	Staying logged in to a website during a session
4	Transport Layer	Ensures complete, correct delivery with error checking	TCP, UDP	Downloading a file (TCP), or watching a live video (UDP)
3	Network Layer	Decides path and handles logical addressing (IP addresses)	IP, ICMP, ARP	Sending data across networks using IP addresses
2	Data Link Layer	Transfers data between devices on same network using MAC address	Ethernet, PPP	Communicating with your router or switch
1	Physical Layer	Transfers raw bits over physical media (cables, signals, etc.)	USB, Bluetooth, Ethernet cable, Fiber Optic	Using LAN cable, Bluetooth headset, or optical connections

TCP/IP Model – 4 Layers (Simplified Explanation)

Layer No.	Layer Name	Purpose (Simple Words)	Example Protocols	Real-Life Example
4	Application Layer	Interacts with user apps (web, email, chat)	HTTP, FTP, SMTP, DNS, DHCP	Browsing, email, file download
3	Transport Layer	Provides reliable/unreliable delivery between devices	TCP, UDP	TCP for file download; UDP for video calls
2	Internet Layer	Handles addressing and routing of data between networks	IP, ICMP, ARP	Sends data to the correct destination using IP address
1	Network Access Layer	Physical transmission of data (hardware level)	Ethernet, Wi-Fi, ARP	Uses LAN, modem, router, or Wi-Fi to send signals

Comparison: OSI vs TCP/IP Models

Feature	OSI Model (7 Layers)	TCP/IP Model (4 Layers)
Full Form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed By	ISO (International Standards Organization)	DARPA (U.S. Department of Defense)
Number of Layers	7	4
Layer Names	Application, Presentation, Session, Transport, Network, Data Link, Physical	Application, Transport, Internet, Network Access
Protocol Dependency	Protocol independent	Protocol specific
Flexibility	Conceptual model – flexible & educational	Practical model – real-world use
Usage in Real Networks	Rarely used directly	Widely used in the Internet
Data Flow Approach	Vertical, strictly layered	More practical, layers can overlap

Use Cases & Real-Time Applications for OSI Model

OSI Layer	Name	Use Case / Scenario	Example Applications / Devices
7	Application Layer	User sending an email, browsing a website	Gmail, Outlook, Google Chrome, WhatsApp, Zoom
6	Presentation Layer	Video streaming with compression, or viewing a secure website (SSL)	SSL/TLS in HTTPS, MP4, JPEG image viewer
5	Session Layer	Login session on an online banking portal	Remote Desktop, NetBIOS, SMB (file sharing)
4	Transport Layer	Downloading a file or live video streaming	TCP (for file transfer), UDP (for live video), Zoom
3	Network Layer	Data routing from India to USA	Routers, IP address (IPv4/IPv6), Google Maps
2	Data Link Layer	Sending data over Wi-Fi within your house	Ethernet switches, MAC address filtering, ARP
1	Physical Layer	Internet through LAN cable or Wi-Fi radio signals	Ethernet cables, fiber optics, Wi-Fi antennas

Use Cases & Real-Time Applications for TCP/IP Model

TCP/IP Layer	Use Case / Scenario	Example Applications / Protocols / Devices
Application Layer	Chatting on WhatsApp, browsing YouTube, using cloud apps	HTTP, HTTPS, FTP, DNS, DHCP, YouTube, Google Meet
Transport Layer	Watching a live match (UDP) vs downloading a PDF (TCP)	TCP (reliable), UDP (fast, less reliable), Netflix, Zoom
Internet Layer	Finding path to send data from one country to another	IP, ICMP (ping), routers, GPS data routing
Network Access Layer	Physically transmitting data from device to router	Ethernet cards, Modem, Wi-Fi adapter, LAN ports

When to Use OSI vs TCP/IP

Scenario	Use OSI Model	Use TCP/IP Model
Teaching/Explaining network concepts	Best for conceptual understanding	Too focused on real-world protocols
Designing network architecture	Helpful for modular design and testing	Used to implement actual protocol stack in networks
Troubleshooting (packet analysis, etc.)	Helps understand where an issue might occur	Use real protocols like TCP, IP, DNS in tools like Wireshark
Real-world internet communication	Theoretical only	TCP/IP runs the Internet & all web services

Latest Applications Examples by Layer (Quick Map)

Activity	Protocols Involved	Related Layer
Video calling (Zoom, Meet)	RTP, UDP, TCP, HTTP	Application, Transport
Online gaming	UDP	Transport
Streaming Netflix/Youtube	HTTP, TCP	Application, Transport
GPS Navigation (Google Maps)	IP, ICMP	Network
Sending files over LAN	SMB, NetBIOS	Session, Data Link
Connecting to Wi-Fi	Ethernet, Wi-Fi	Data Link, Physical

What is UDP? (User Datagram Protocol)

UDP is a **Transport Layer protocol** used for sending data quickly over the network **without establishing a connection**. It is called "**connectionless**" and "**unreliable**", but **very fast**.

UDP Full Form: User Datagram Protocol

Key Features of UDP:

Feature	Description
Connectionless	No need to set up a connection before sending data.
Unreliable	No guarantee that the data will reach the receiver.
No Acknowledgment	Sender does not wait for a confirmation.
Fast Transmission	Less overhead, suitable for time-sensitive tasks.

Message-Based	Data is sent as individual packets called datagrams .
No Congestion Control	Does not reduce speed even when the network is busy.
No Flow Control	Sender sends data at its own speed without checking receiver speed.

How UDP Works (Simple Steps)

1. **Sender creates a datagram** with data and destination IP + port.
2. The datagram is **sent directly** to the receiver (no handshake).
3. Receiver receives it (if it arrives), but **does not send any acknowledgment**.
4. If packet is lost, it is simply lost — **no retransmission**.

UDP Packet Structure

A UDP packet is small and lightweight. It has:

- **Source Port**
- **Destination Port**
- **Length** (of the packet)
- **Checksum** (error detection only, not correction)
- **Data Payload**

The **header** is only **8 bytes**, making it much lighter than TCP's 20-byte header.

When to Use UDP (Use Cases)

Use Case	Reason
Video Streaming (YouTube Live)	Small loss is okay; speed is more important.
Voice Calls (VoIP like WhatsApp/Skype)	Low delay is important; packets must arrive quickly.
Online Gaming	Fast response needed; slight data loss doesn't affect experience.
DNS Queries	Quick single-message request-response; speed matters.
Broadcast/Multicast Traffic	UDP supports broadcasting to multiple users.

Advantages of UDP

- Very **fast** (no connection setup).
- **Lightweight** header (less bandwidth).
- Good for **real-time applications**.
- Supports **multicast/broadcast**.

Disadvantages of UDP

- **No error recovery** (packet loss is not fixed).
- **No ordering** (packets may arrive out of order).
- **No congestion control** (can overload the network).
- **Not suitable for critical data** (like emails, files, banking).

UDP vs TCP – Key Differences

Feature	UDP	TCP
Connection	No (connectionless)	Yes (connection-oriented)
Reliability	Unreliable (no retransmission)	Reliable (ensures delivery)
Speed	Fast	Slower due to acknowledgments
Data Order	No guarantee	Guaranteed
Use Case	Streaming, VoIP, Gaming	File transfer, Web browsing, Emails
Header Size	8 bytes	20 bytes
Error Handling	Only checksum (detection only)	Full error checking and retransmission

Real-World Examples Using UDP

Application	Why it uses UDP
WhatsApp Audio Call	Real-time, can't wait for retransmission
Netflix / YouTube Live	Stream keeps playing, even if some data lost
Online Games (PUBG)	Lag-free gaming; packet loss is tolerable
DNS Lookup	Quick response needed
DHCP	Assigns IPs quickly without connection

Critique of OSI Model (Open Systems Interconnection Model)

The OSI model is a **theoretical framework** that divides networking into **7 layers** to help standardize communications.

Advantages of OSI Model

Point	Explanation
1. Standardization	Helps vendors and developers understand network structure and build compatible systems.
2. Layered Approach	Divides the job into smaller, manageable parts (each layer has a clear task).
3. Interoperability	Makes it easier for different devices and software to work together.
4. Troubleshooting	Easier to find and fix network problems by isolating them layer-by-layer.
5. Protocol Independence	New protocols can be developed and plugged into a layer without changing others.

Disadvantages / Criticism of OSI Model

Point	Explanation
1. Too Complex for Practical Use	Some layers are not clearly separate in real life (e.g., session and presentation).
2. Rarely Implemented Fully	No real-world protocol stack perfectly matches all 7 layers.
3. Slow Standardization	OSI protocols were developed too slowly and lost adoption to TCP/IP.
4. Redundant Functions	Some layers perform overlapping tasks (e.g., error control in both transport and data link).
5. Lack of Real-World Examples	Mostly used for teaching, not actual implementation.

Critique of TCP/IP Model (Transmission Control Protocol / Internet Protocol)

TCP/IP is the **actual protocol suite** used on the Internet. It has **4 layers**: Application, Transport, Internet, and Network Access.

Advantages of TCP/IP Model

Point	Explanation
1. Widely Used and Practical	It's the protocol that runs the internet, tested and trusted globally.
2. Simple and Efficient	Fewer layers, faster development, easy to understand.
3. Robust & Scalable	Can handle millions of users and various applications.
4. Open and Flexible	Can support many routing and transport technologies.
5. Reliable	Built-in mechanisms for packet ordering, error checking, and retransmission (TCP).

Disadvantages / Criticism of TCP/IP Model

Point	Explanation
1. No Clear Separation of Layers	Mixing of functions (e.g., Application layer does work of OSI's top 3 layers).
2. Poorly Defined Interfaces	Not as modular as OSI; tough to update/replace individual parts.
3. No Formal Standard for Services	OSI defines service, interface, and protocols — TCP/IP does not.
4. No Support for Future Technologies (Initially)	Was not designed with things like mobile networks or streaming in mind.
5. Focus on Connection-Oriented Services	Assumes the use of reliable protocols like TCP — UDP is less considered.

Comparison: OSI vs TCP/IP Models

Feature	OSI Model	TCP/IP Model
Type	Conceptual/theoretical model	Practical protocol suite
Layers	7	4 (sometimes shown as 5)
Development	By ISO	By U.S. Department of Defense (DARPA)
Used in Practice	Rarely used (mostly educational)	Widely used (Internet backbone)

Protocols Defined	No real protocols (mostly guidelines)	Real protocols like TCP, IP, FTP, etc.
Layer Separation	Clear and distinct	Overlapping and combined functions
Flexibility	Very flexible but impractical	Less flexible but highly practical
Error Handling	Error handling at multiple layers	Primarily in Transport layer (TCP)

Real-World Use Case:

Real-World Scenario	Model Used
Internet browsing	TCP/IP (HTTP over TCP/IP)
Learning & network design	OSI Model (Conceptual clarity)
Network troubleshooting	OSI (Layer-by-layer debugging) + TCP/IP (real protocols)
Protocol stack in Linux	TCP/IP stack
Teaching networking	OSI model for theory, TCP/IP for implementation

Criticism of TCP/IP Protocols – Explained Simply

1. Lack of Security

- **Issue:** TCP/IP was never designed with **security** in mind.
- **Why it's a problem:** There is **no built-in encryption, authentication, or integrity checks**.
- **Example:** Attacks like IP spoofing, TCP hijacking, and DDoS (Distributed Denial of Service) are common due to weak native protection.

2. Poor Quality of Service (QoS) Support

- **Issue:** No mechanism to guarantee bandwidth, delay, or priority handling.
- **Why it's a problem:** In real-time apps like video conferencing or online gaming, **delay and jitter** hurt user experience.
- **Example:** VoIP and Zoom calls may suffer without QoS.

3. Limited Support for Mobile and Wireless Networks

- **Issue:** Originally built for static devices, not for modern **mobile, wireless, or IoT networks**.
- **Why it's a problem:** Mobile IP was introduced as an afterthought and doesn't handle roaming well.
- **Example:** IP address changes when a mobile device switches networks.

4. Address Exhaustion (IPv4)

- **Issue:** IPv4 supports only ~4.3 billion unique addresses.
- **Why it's a problem:** Explosion of devices (smartphones, sensors, IoT) has caused IPv4 to run out.
- **Example:** Workarounds like NAT (Network Address Translation) add complexity and break end-to-end connectivity.

5. Complex Configuration

- **Issue:** Manual setup of IP addresses, subnet masks, and routing can be confusing.
- **Why it's a problem:** Misconfiguration leads to network failures.
- **Example:** DHCP (Dynamic Host Configuration Protocol) was introduced later to solve this, but the protocol stack itself lacks automation.

6. TCP Overhead

- **Issue:** TCP provides reliability, but with **a lot of control mechanisms** (handshake, flow control, congestion control).
- **Why it's a problem:** For some apps, this is **too heavy** and reduces performance.
- **Example:** Online games or live video often prefer **UDP**, which is faster but less reliable.

7. Scalability and Routing Complexity

- **Issue:** Routing protocols like BGP (Border Gateway Protocol) and OSPF are **complex and vulnerable**.
- **Why it's a problem:** Large networks (like the Internet) can be misconfigured, hacked, or cause outages.
- **Example:** BGP hijacking can redirect internet traffic maliciously.

8. Lack of Layer Independence

- **Issue:** Application layer protocols are closely tied to transport layer (like HTTP to TCP).

- **Why it's a problem:** It's harder to innovate or switch layers.
- **Example:** HTTP/3 uses UDP with QUIC to bypass TCP's slowness, but it's not well supported everywhere yet.

9. No Built-in Resource Management

- **Issue:** TCP/IP doesn't control bandwidth or traffic shaping by default.
- **Why it's a problem:** Network congestion is hard to manage in high-traffic systems.
- **Example:** TCP slow start is helpful, but doesn't handle modern cloud traffic well.

Criticism of OSI Protocols (Not Just the Model)

While the **OSI model** is useful for learning and structuring, the **OSI protocols** (those created to implement OSI layers) faced many issues and were not widely adopted. Here's why:

1. Too Complex and Heavyweight

- **Problem:** OSI protocols (like CLNP for network layer, TP4 for transport) were very detailed and had **too many features**.
- **Why it failed:** Complexity slowed implementation and performance.
- **Example:** TCP/IP became more popular because it was simpler and "just worked."

2. Slow Development and Standardization

- **Problem:** The OSI protocols took **too long to finalize** (bureaucratic process).
- **Why it failed:** By the time OSI protocols were standardized, TCP/IP was already widely used.
- **Example:** TCP/IP was running the ARPANET and the early Internet before OSI protocols even rolled out.

3. Lack of Open Implementation

- **Problem:** OSI protocols were not freely available in open-source or public implementations.
- **Why it failed:** Developers had a harder time experimenting or building products.
- **Example:** TCP/IP was implemented in open-source systems like BSD Unix, accelerating adoption.

4. Too Much Layer Rigidity

- **Problem:** Each OSI protocol was designed to match *exactly one* OSI layer.
- **Why it failed:** In real systems, some protocols span multiple layers.

- **Example:** TCP/IP protocols like TCP (layer 4) and HTTP (layer 7) interact freely; OSI discouraged that.

5. Vendor-Driven Rather than Community-Driven

- **Problem:** OSI was mostly backed by governments and large telecom vendors (ISO, ITU-T).
- **Why it failed:** It didn't adapt to practical developer needs or real-world internet usage.
- **Example:** TCP/IP was community-driven and quickly evolved to meet needs.

6. Duplicate Functionality Across Layers

- **Problem:** OSI protocols had overlapping functions across layers.
- **Why it failed:** This created confusion and inefficiency.
- **Example:** Both the Transport (TP) and Session (SP) layers handled error recovery in different ways

7. Poor Performance

- **Problem:** OSI protocols were slow and heavy due to too many features.
- **Why it failed:** They couldn't match the performance of leaner TCP/IP protocols.
- **Example:** TP4 (Transport Protocol class 4) was slower than TCP in practice.